

专题12. 反常积分计算的解题方法

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \frac{x}{a} + C$$

一. 积分区间内部如果存在瑕点, 则需要拆区间.

例1. $\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{|x-x^2|}} dx = \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{|x(1-x)|}} dx$

$$x-x^2 = \frac{1}{4} = \left(x-\frac{1}{2}\right)^2 \quad x^2-x = \left(x-\frac{1}{2}\right)^2 - \frac{1}{4}$$

解: $I = \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{x(1-x)}} dx + \int_1^{\frac{3}{2}} \frac{1}{\sqrt{x(1-x)}} dx = \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{\frac{1}{4} - \left(x-\frac{1}{2}\right)^2}} d\left(x-\frac{1}{2}\right) + \int_1^{\frac{3}{2}} \frac{1}{\sqrt{\left(x-\frac{1}{2}\right)^2 - \frac{1}{4}}} d\left(x-\frac{1}{2}\right)$

$$= \arcsin \frac{x-\frac{1}{2}}{\frac{1}{2}} \Big|_{\frac{1}{2}}^1 + \ln \left[\left(x-\frac{1}{2}\right) + \sqrt{\left(x-\frac{1}{2}\right)^2 - \frac{1}{4}} \right] \Big|_1^{\frac{3}{2}} = \frac{\pi}{2} + \ln \frac{1+\frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

$$\ln^a - \ln^b = \ln \frac{a}{b}$$

$$= \frac{\pi}{2} + \ln(2+\sqrt{3})$$

$$\int \frac{1}{t(t+1)} dt = \int \frac{1}{t} dt - \int \frac{1}{t+1} dt$$

二. 将反常积分裂开成两个积分时, 也需要考虑每个积分的敛散性.

例2. $\int_1^{+\infty} \frac{1}{x(x^2+1)} dx = \frac{1}{2} \int_1^{+\infty} \frac{1}{x^2(x^2+1)} dx \stackrel{x=t}{=} \frac{1}{2} \int_1^{+\infty} \frac{1}{t^2(t+1)} dt \neq \frac{1}{2} \left[\int_1^{+\infty} \frac{1}{t^2} dt + \int_1^{+\infty} \frac{1}{t+1} dt \right] \times \times \times \ln t \Big|_1^{+\infty}$

解: $I = \dots = \frac{1}{2} \int_1^{+\infty} \frac{1}{t(t+1)} dt = \frac{1}{2} \int_1^{+\infty} \left(\frac{1}{t} - \frac{1}{t+1} \right) dt = \frac{1}{2} \ln \frac{t}{t+1} \Big|_1^{+\infty} = \frac{1}{2} [\ln 1 - \ln \frac{1}{2}] = \frac{1}{2} \ln 2$
 $= \frac{1}{2} \cdot (\ln t - \ln(t+1)) \Big|_1^{+\infty} \quad -\ln \frac{1}{2} = \ln 2$

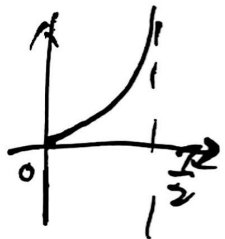
③ 反常积分的分部积分, 也需要考虑每一项的敛散性.

例3. $\int_0^{+\infty} \frac{x e^{-x}}{(1+e^{-x})^2} dx$ $\star\star\star$. $(e^x)^2 \rightarrow \infty$ $\frac{e^x}{e^x+1} \xrightarrow{x \rightarrow +\infty} 1$

[解]: $I = - \int_0^{+\infty} \frac{x}{(1+e^{-x})^2} d(e^{-x}+1) = \int_0^{+\infty} x d \frac{1}{e^x+1} = \frac{x}{e^x+1} \Big|_0^{+\infty} - \int_0^{+\infty} \frac{1}{e^x+1} dx$ (X)

[正解]: $I = \int_0^{+\infty} x d(\frac{1}{e^x+1} - 1) = \int_0^{+\infty} x d \frac{e^{-x}}{e^x+1} = - \int_0^{+\infty} x d \frac{1}{e^x+1} = - \frac{x}{e^x+1} \Big|_0^{+\infty} + \int_0^{+\infty} \frac{1}{e^x+1} dx$
 $= \int_0^{+\infty} \frac{1}{e^x(e^x+1)} dx = \ln \frac{e^x}{e^x+1} \Big|_0^{+\infty} = 0 - \ln \frac{1}{2} = \ln 2 \checkmark$

[证]: $I = \int_0^{+\infty} \frac{x e^x}{(e^x+1)^2} dx = - \int_0^{+\infty} x d \frac{1}{e^x+1} = \dots = \ln 2 \checkmark$
 $-\frac{1}{t^2} dt$ $x = \frac{1}{t}$ $t = \frac{1}{x}$
 $x: 1 \rightarrow +\infty$ $t: 1 \rightarrow 0$



四. $(0, +\infty)$ 上的区间再收公式 $\begin{cases} (I_1 =) (0, +\infty) \text{ 拆成 } (0, 1) \text{ 和 } (1, +\infty) \text{ 对 } (1, +\infty) \text{ 上的积分用倒代换} \\ (I_2 =) \text{ 直接倒代换} \end{cases}$

例4. $\int_0^{+\infty} \frac{1}{(1+x^2)(1+x^a)} dx$
[解]: 令 $x = \frac{1}{t} \Rightarrow I = \int_{+\infty}^0 \frac{1}{(1+\frac{1}{t^2})(1+\frac{1}{t^a})} \cdot (-\frac{1}{t^2}) dt = \int_0^{+\infty} \frac{t^a}{(1+t^2)(1+t^a)} dt = \frac{1}{2} \int_0^{+\infty} \frac{1+t^a}{(1+t^2)(1+t^a)} dt = \frac{1}{2} \int_0^{+\infty} \frac{1}{1+t^2} dt = \frac{1}{2} \arctan t \Big|_0^{+\infty} = \frac{\pi}{4} \checkmark$

[另解]: 令 $x = \tan t \Rightarrow I = \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2 t}{\sec^2 t (1+\tan^a t)} dt = \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^a t}{\cos^a t + \sin^a t} dt$ (区间再收)
 $\int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^a t + \sin^a t - \sin^a t}{\cos^a t + \sin^a t} dt = \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dt = \frac{\pi}{2} - \frac{\pi}{2} = 0$ (Note: The original image has a different result for this method, likely due to a typo in the limits or the integrand. The correct result for the integral from 0 to infinity is pi/4.)

例5. $\int_0^{+\infty} \frac{\ln x}{1+x^2} dx = ?$

[解]: $\frac{1}{x} = t \Rightarrow I = \int_{+\infty}^0 \frac{-\ln t}{1+\frac{1}{t^2}} \cdot (-\frac{1}{t^2}) dt = \int_0^{+\infty} \frac{\ln t}{t^2+1} dt = -I \Rightarrow I=0$

[另解]: $\frac{1}{x} = \tan t \Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\ln \tan t}{\sec^2 t} \cdot \sec^2 t dt = \int_0^{\frac{\pi}{2}} \ln \frac{\sin t}{\cos t} dt = \int_0^{\frac{\pi}{2}} \ln \sin t dt$

$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \ln \frac{\sin t}{\cos t} \frac{\cos t}{\sin t} dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} 0 dt = 0$ ✓

(例题) $\int_0^{+\infty} \frac{\ln x}{a^2+x^2} dx \quad (a>0)$

[解]: 代换: $\frac{1}{x} = at \Rightarrow I = \int_0^{+\infty} \frac{\ln a + \ln t}{a^2 + a^2 t^2} a dt = \frac{1}{a} \left[\int_0^{+\infty} \frac{\ln a}{1+t^2} dt + \int_0^{+\infty} \frac{\ln t}{1+t^2} dt \right]$

[另解]: 三角换元: $\frac{1}{x} = a \tan t \Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\ln a + \ln \tan t}{a^2 \sec^2 t} \cdot a \sec^2 t dt = \frac{\pi}{2} \cdot \frac{\ln a}{a} + \left(\int_0^{\frac{\pi}{2}} \ln \tan t dt \right) \cdot \frac{1}{a}$
 $= \frac{\pi}{2} \cdot \frac{\ln a}{a}$ ✓

[另解]: 倒代换: $\frac{1}{x} = \frac{a}{t}$

$a^2 + \frac{a^2}{t^2}$

$a^2 + x^2 \rightarrow x = a \tan t$
 $a^2 - x^2 \rightarrow x = a \sin t$

$\frac{\ln a}{a} \cdot \frac{\pi}{2}$ ✓

例6. $\int_0^{+\infty} \frac{\ln x}{1+x+x^2} dx$

[解]: $\frac{1}{x} x = \frac{1}{t} \Rightarrow I = \int_0^{+\infty} \frac{-\ln t}{1+\frac{1}{t}+\frac{1}{t^2}} \cdot (-\frac{1}{t^2}) dt = -\int_0^{+\infty} \frac{\ln t}{t^2+t+1} dt = -I \Rightarrow I=0$ ✓

变题: $\int_0^{+\infty} \frac{\ln(1-x+x^2)}{x(1+x^2)} dx = I$

[解]: $\frac{1}{x} x = \frac{1}{t} \Rightarrow I = \int_0^{+\infty} \frac{\ln(1-\frac{1}{t}+\frac{1}{t^2})}{(1+\frac{1}{t^2}) \cdot (-\frac{1}{t^2})} \cdot (-\frac{1}{t^2}) dt$
 $\rightarrow \ln \frac{t^2-t+1}{t^2} \quad \ln \frac{a}{b} = \ln a - \ln b$
 $= -\int_0^{+\infty} \frac{\ln(t^2-t+1) - 2\ln t}{(t^2+1)\ln t} dt = -I + 2 \cdot \int_0^{+\infty} \frac{1}{t^2+1} dt$

$\Rightarrow I = \int_0^{+\infty} \frac{1}{t^2+1} dt = \arctan t \Big|_0^{+\infty} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$ ✓